

COBALT: Active Learning Loop for Categorical Structural Optimization

Discrete Manifold Anchoring

Physical Cross-Section Catalog $\mathcal{C}$



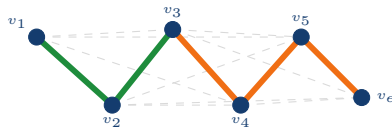
GP-LVM Φ



Anchor to discrete grid Ω_D ;
reject continuous relaxation

Random Decomposition

Categorical Variables

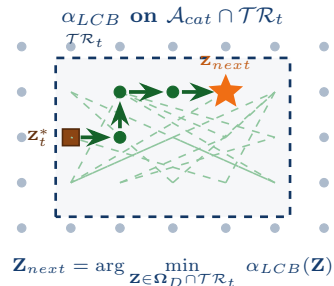


Random spanning tree
 $\mathcal{E}_t \sim \text{Uniform}(\mathcal{G}_{\text{tree}})$

$$k_{\mathcal{G}_t}(\mathbf{Z}, \mathbf{Z}') = \sum_{(u,v) \in \mathcal{E}_t} k_{uv}(\mathbf{y}_u, \mathbf{y}_v; \mathbf{y}'_u, \mathbf{y}'_v)$$

Info-gain bound $\gamma_T = \mathcal{O}(e(\log T)^3)$

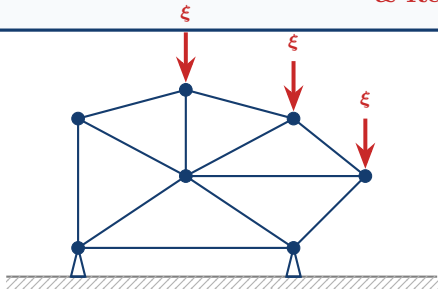
Trust-Region Acquisition



Pure Discrete Search
Zero decoding failures

Update SAAS-GP
& Resample tree $\mathcal{G}_t$

Assemble Physical
Structure $\mathbf{Z}_{\text{next}}$



Stochastic Structural MC-FEA Oracle

$$\mathcal{J}_{\text{robust}}(\mathbf{Z}) = \mathbb{E}_{\xi}[J(\mathbf{Z}, \xi)] + \gamma \sqrt{\mathbb{V}_{\xi}[J(\mathbf{Z}, \xi)]}$$

$$y_{\text{obs}} = \mathcal{J}_{\text{robust}}(\mathbf{Z}_{\text{next}}) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_n^2)$$

Evaluates exact discrete structure under aleatoric uncertainty ξ
to close the active learning loop.